ZDC autoresearch, Phase 4: the Λ bar fight

Maximize $\Lambda \to n\pi^0$ reconstructed in EICrecon per thrown Λ — no gates, fair-play rules

Started and closed 2026-11-08 · one slide per try · rebuilt from the run's result files

Phase 4 result — X9 against the start and the ceilings

7.436 %
X9, mean over 100–275 GeV

14.101 %
X9, at 275 GeV

14.1 / 27.8 %
ceiling, mean / 275 GeV (414)

S4a_tm per thrown Λ → nπ⁰	mean 100–275 GeV	at 275 GeV
R1 (PR 3064), the start	0.404 %	0.832 %
X9, the final row	7.436 %	14.101 %
ceiling (perfect separation, P1b resolution)	14.1 %	27.8 %
acceptance (all three daughters reach the ZDC)	16.91 %	32.64 %

How it was earned

- 9 tries: 8 accepted, 1 rejected (X6), 0 foul
- fair play PASS on every try (460)
- held-out D4h reproduces at both checkpoints (z +36, +31)
- determinism repeats identical
- commits local on zdc-autoresearch/phase4; never pushed

18× the start on the mean and 17× at 275 GeV; 43 % of the Λ in the acceptance at 275 GeV are found.

Phase 4 score — S4a_tm per thrown $\Lambda \to n\pi^0$ and the starting rows

0.404 %

R1 (PR 3064), mean 100–275 GeV

0.832 %

R1 (PR 3064), at 275 GeV

14.1 / 27.8 %

row	in ElCrecon	offline, tagged + refit	offline, truth-identified + refit
R1_pr3064	0.404 % / 0.832 %	1.643 % / 3.066 %	3.419 % / 5.689 %
P1d_R1_nlr1	0.390 % / 0.947 %	1.697 % / 3.508 %	4.019 % / 6.984 %

Each cell: mean over 100–275 GeV / value at 275 GeV, per thrown $\Lambda \to n\pi^0$ (headline.csv). Offline rows run the stable-vertex chain on the row's own clusters (415); the truth-identified row picks the three clusters by truth.

Selector (D-default-42)

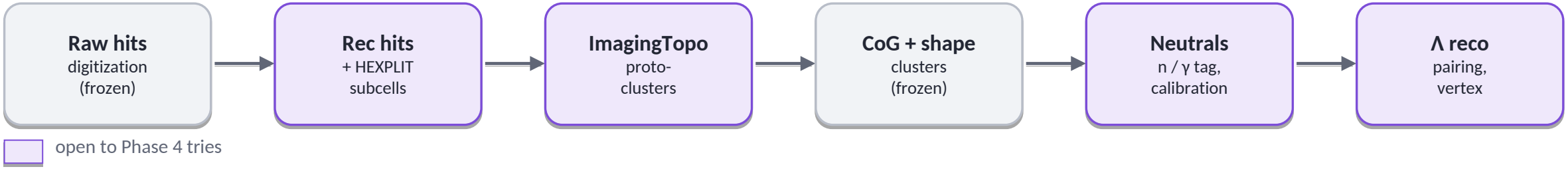
- ACCEPTED when the pooled McNemar z over the eight D4 momenta is at least 3 against the current best on identical events, and the mean per thrown rises

Fair play — 460 checks F1–F3 and fails on every damaged control

row	role	F1 truth-blind	F2 raw hits = R0	F3 honest candidates
R0_stock	reference	PASS	PASS	PASS
R1_pr3064	reference	PASS	PASS	PASS
R2_pr3114	reference	PASS	PASS	PASS
R3_pr3063	reference	PASS	PASS	PASS
L01_box3114	reference	PASS	PASS	PASS
Rt_truthchain	truth chain beside the scored one	FAIL	PASS	PASS
CTL_F1_inplace	truth clusters feed the scored chain	FAIL	PASS	—
CTL_F1_rewire	InputTags rewires Λ to truth neutrals	FAIL	PASS	PASS
CTL_F2_digi	ZDC pedestal σ 2 → 8 ADC	PASS	FAIL	PASS
CTL_F3_dup	two candidates; photon links all clusters	PASS	PASS	FAIL

F1: static walk of the factory registrations from ReconstructedLambdas to raw hits, the patch, the parameters, and the factories that run (janadot). F2: digitization untouched, raw hits byte-identical to R0. F3: one Λ per event, n γ γ daughters, disjoint clusters, E_Λ = Σ daughters, no mass constraint.

ZDC Λ chain in EICrecon — the stages a try may change



Frozen (fair-play files)

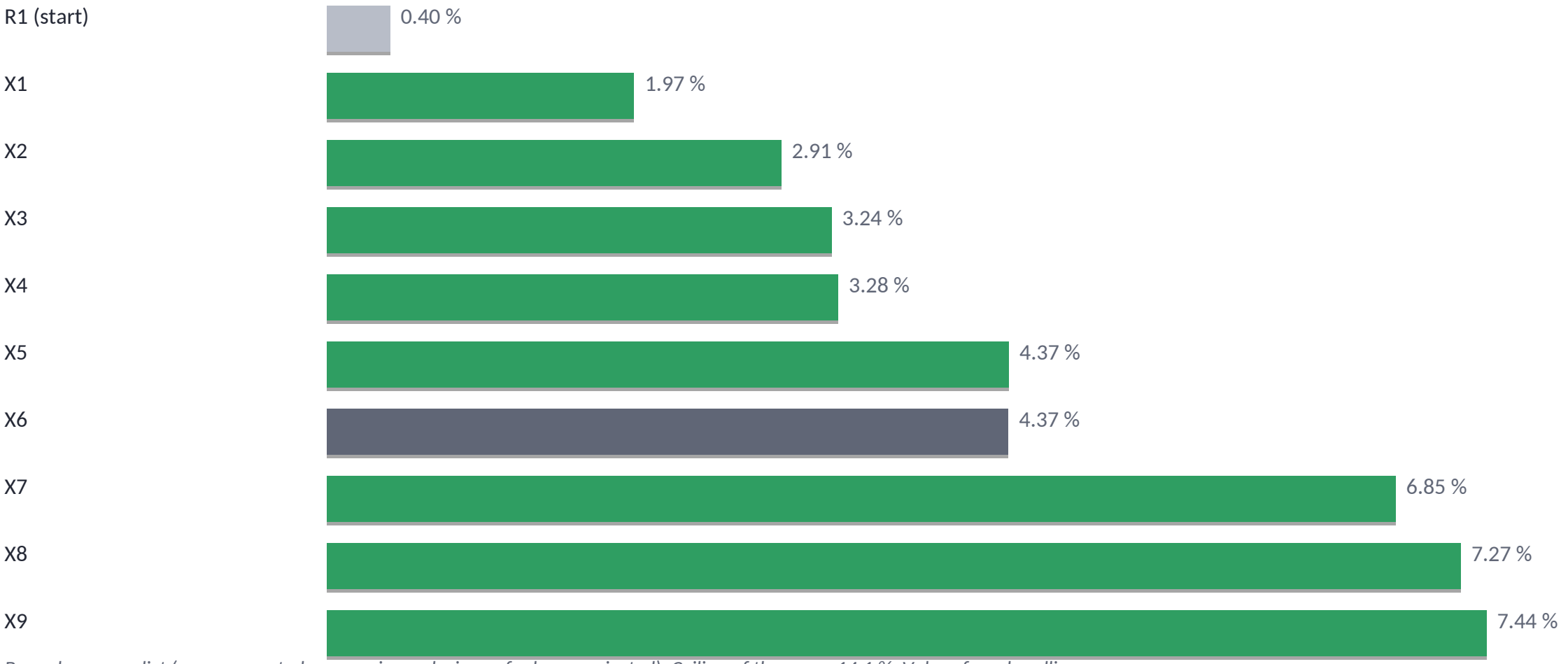
- digitization: the raw hits are the data (F2)
- the CoG step and the cluster shapes: they write the cluster ↔ MC associations the scorer reads (F4)
- the scorer: 410 truth match, ±30 MeV window, purity 0.8, the D4 / D4h samples

Open

- hit reconstruction and HEXPLIT subcells
- clustering (ImagingTopoCluster, Island, new algorithms that feed the CoG)
- the n / γ tag and the calibration (fitted on the D2 / D3 guns only)
- the Λ reconstruction: pairing, vertex, ranking — one candidate per event

Every try below highlights its stages on this strip.

Score per try — S4a_tm per thrown $\Lambda \to n\pi^0$, mean over 100–275 GeV



Bar colour = verdict (green accepted, orange inconclusive or foul, grey rejected). Ceiling of the mean 14.1 %. Values from headline.csv.

X1 — the offline Λ chain built into EICrecon

Change

- R1 clustering (PR 3064) with neighbourLayersRange 1 (P1d)
- P1d's calibration refit on the D2 / D3 guns: photon 3.90·E^0.876, neutron 8.06·E^0.700 (PR 3064's laws overestimate photons 4.4× and neutrons 1.7× at 275 GeV)
- every γγ pair with every neutron goes through the vertex fit; the best fitted π^0 and Λ masses win
- stable vertex: depth by bisection along the momentum-weighted slope (B9d)
- neutron position at its leader cluster (B9e)

Prediction (written before launch)

- 1.5 % mean, 3.0 % at 275 GeV per thrown (R1's offline tagged refit: 1.64 / 3.07 %; in-chain pairing loses events where truth chose the triplet offline)

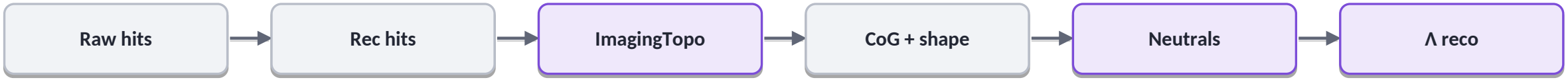
1.970 %
mean per thrown

4.309 %
at 275 GeV

current best R1_pr3064: 0.404 % mean, 0.832 % at 275 GeV · ceilings 14.1 % mean, 27.8 % at 275 GeV

paired test: gained 1325, lost 154, z +30.45

ACCEPTED | **fair play PASS**



purple: stages this try changes

X1 — Λ reconstruction flow at the pin and in X1

Pin (stock)

γγ pairs: π^0 mass from origin-pointing momenta
cut at ±10 % of m_{π^0}

↓

n γγ triplets: Λ mass from origin-pointing momenta
cut at ±10 % of m_Λ

↓

rank: ZDC first, then mass χ^2, p_z, E
one triplet

↓

vertex: iterate along the Λ direction
neutron at its first linked cluster

↓

final cut ±10 % of m_Λ
one candidate or none

X1

every γγ pair with every neutron
no cut before the vertex

↓

vertex: bisection on the γγ opening angle
along the momentum-weighted slope (B9d)

↓

neutron at its leader cluster
reference point (B9e)

↓

rank by fitted π^0 and Λ masses
discrete choice only; energies untouched

↓

one candidate per event
the scorer applies ±30 MeV

A decay 10–30 m downstream makes origin-pointing masses wrong; X1 computes every mass after the vertex fit.

X2 — the photon pair chosen by the Λ hypothesis in the ZDC neutrals

Change

- X1 finds the true γγ pair in about half of the events where one exists (471 on X1's events)
- tagged clusters the pair leaves out are mostly neutron fragments: their energy belongs to the neutron
- for every pair of tagged clusters: neutron = every other cluster (energy sum, leader position)
- rank by fitted π^0 and Λ mass χ^2 plus the Λ decay-depth density ($c\tau$ = 78.9 mm)
- only the chosen pair is written as photons; energies are never rescaled

Prediction (written before launch)

- 2.84 % mean, 5.45 % at 275 GeV per thrown (offline emulation of the rule on X1's events, which reproduces X1's own count at every momentum)

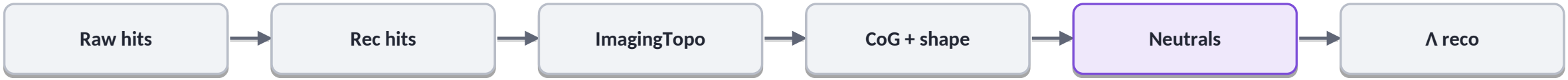
2.913 %
mean per thrown

5.500 %
at 275 GeV

current best X1_offline_chain: 1.970 % mean, 4.309 % at 275 GeV · ceilings 14.1 % mean, 27.8 % at 275 GeV

paired test: gained 1433, lost 416, z +23.65

ACCEPTED **fair play PASS**



purple: stages this try changes

X2 — photon choice in the ZDC neutrals at X1 and in X2

X1

ZDC clusters → shape box tags photons
about 5–9 tagged clusters per event

↓

every tagged cluster becomes a photon
neutron fragments included

↓

neutron = untagged clusters only
fragment energy missing

↓

Λ reconstruction ranks all pairs
fitted π^0 and Λ masses

↓

one candidate per event

X2

ZDC clusters → shape box tags candidates
unchanged

↓

each pair of candidates: neutron = all other clusters
energy sum, leader position

↓

stable-vertex fit per pair
π^0 constraint, as LambdaReconstruction

↓

rank: π^0 and Λ mass χ^2 + Λ decay-depth density
discrete choice; energies untouched

↓

only the chosen pair written as photons
the rest stays with the neutron

The neutron of each hypothesis holds every cluster the pair leaves out, so the true pair also gets the right neutron energy.

X3 — the Λ-pairing ranking weights

Change

- X2 ranks pairs by $(\Delta m_{\pi^0} / 0.1\, m_{\pi^0})^2 + (\Delta m_\Lambda / \sigma_\Lambda)^2 + w \cdot$ depth term
- X2: σ_Λ = 111 MeV (0.1 m_Λ), w = 1; X3: σ_Λ = 40 MeV (about the mass resolution), w = 1.5
- chosen at the centre of a plateau in a 32-point grid on X1's events (471): +12 % at every momentum
- both are parameters now: lambdaPairingLambdaSigma, lambdaPairingDepthWeight
- D4h at checkpoint 1 tests the D4-chosen weights

Prediction (written before launch)

- 3.24 % mean, 5.98 % at 275 GeV per thrown (X2 scaled per momentum by the grid)

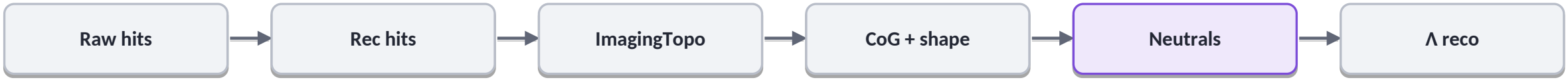
3.236 %
mean per thrown

5.957 %
at 275 GeV

current best X2_lambda_pairing: 2.913 % mean, 5.500 % at 275 GeV · ceilings 14.1 % mean, 27.8 % at 275 GeV

paired test: gained 384, lost 88, z +13.62

ACCEPTED **fair play PASS**



purple: stages this try changes

X4 — a depth limit on the photon tag (as set, 1141 mm)

Change

- true-photon clusters sit at 126 mm median depth past the ZDC face, other tagged clusters at 280 mm
- intended: photon candidates within 225 mm of the face (gammaZMaxOffset 400 → 225)
- the offset is added to HcalFarForwardZDC_SiPMonTile_r_pos, the ZDC centre (face + 916 mm)
- so the limit landed at 1141 mm (stock: 1316 mm, no effective depth cut)
- the prediction failed; X5 sets the limit at the face as intended

Prediction (written before launch)

- 4.31 % mean, 7.96 % at 275 GeV per thrown (471 at 225 mm past the face) — failed: mechanism mis-set

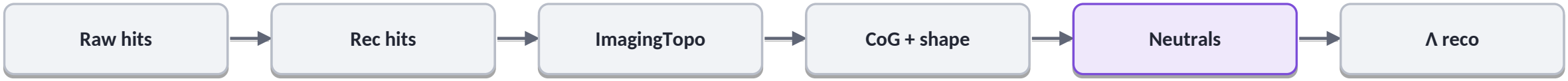
3.278 %
mean per thrown

6.055 %
at 275 GeV

current best X3_pairing_weights: 3.236 % mean, 5.957 % at 275 GeV · ceilings 14.1 % mean, 27.8 % at 275 GeV

paired test: gained 32, lost 0, z +5.66

ACCEPTED **fair play PASS**



purple: stages this try changes

X5 — photon candidates within 225 mm of the ZDC face

Change

- gammaZMaxOffset = 225 − 916.4 = −691.4 mm, so the depth limit sits 225 mm past the front face
- fewer neutron fragments compete as photon candidates in the Λ pairing
- the excluded clusters join the neutron, whose energy becomes more complete
- chosen at the centre of the 200–225 mm plateau of 471 on X1's events
- a parameter on X3's build; no new code

Prediction (written before launch)

- 4.31 % mean, 7.96 % at 275 GeV per thrown (471, depth 225 mm, neutron holding every non-pair cluster)

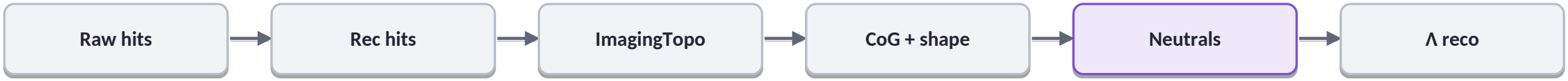
4.373 %
mean per thrown

8.030 %
at 275 GeV

current best X4_photon_depth: 3.278 % mean, 6.055 % at 275 GeV · ceilings 14.1 % mean, 27.8 % at 275 GeV

paired test: gained 972, lost 28, z +29.85

ACCEPTED **fair play PASS**



purple: stages this try changes

X6 — the vertex range ending at the ZDC front face

Change

- LambdaReconstruction and the pairing searched the decay vertex up to the ZDC centre (front + 916 mm)
- X6 ends the vertex range and the decay-depth density at the front face (35.8 m)
- the photon depth limit stays 225 mm past the face
- three parameters on X5's code; no new code

Prediction (written before launch)

- a small gain: 4.40 % mean, 8.1 % at 275 GeV per thrown

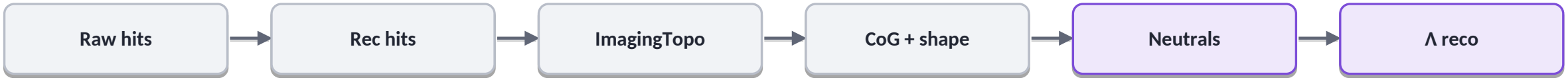
4.370 %
mean per thrown

7.997 %
at 275 GeV

current best X5_photon_depth_face: 4.373 % mean, 8.030 % at 275 GeV · ceilings 14.1 % mean, 27.8 % at 275 GeV

paired test: gained 5, lost 6, z -0.30

REJECTED | fair play PASS



purple: stages this try changes

X7 — photon cores split from the ZDC clusters, lower cluster floor

Change

- with X5, only 28 % of D4 events have three clean clusters; photons sit inside the neutron's cluster
- each imaging cluster is split between its shallow cores (≥ 5 MeV, first 4 layers, 50 mm apart) and one deep region
- hits join the region of their most energetic neighbour; deep showers stay whole
- cluster floor 50 → 20 MeV recovers low-energy photons with no cluster
- calibration refit on D2 / D3 for this clustering (photon 4.50·E^0.848, neutron 6.21·E^0.740)

Prediction (written before launch)

- 6.4 % mean, 11.8 % at 275 GeV per thrown (two-file scans: +46–48 %)

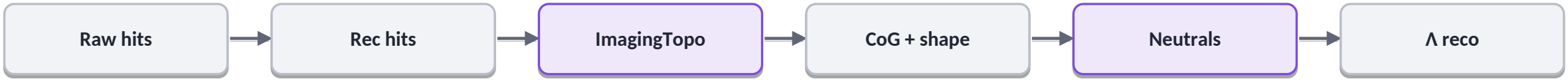
6.853 %
mean per thrown

12.942 %
at 275 GeV

current best X5_photon_depth_face: 4.373 % mean, 8.030 % at 275 GeV · ceilings 14.1 % mean, 27.8 % at 275 GeV

paired test: gained 2827, lost 308, z +44.99

ACCEPTED **fair play PASS**



purple: stages this try changes

X7 — ZDC imaging clusters at X5 and in X7

X5

HEXPLIT subcells → connected groups
neighbours in XY and ±1 layer

↓

a group survives at ≥ 50 MeV
low-energy photons often lost

↓

one cluster per group
a photon touching the neutron shower merges into it

↓

photon tag: shape box, depth ≤ 225 mm

↓

Λ pairing (X2–X5)

X7

HEXPLIT subcells → connected groups
unchanged

↓

a group survives at ≥ 20 MeV
low-energy photons kept

↓

cores: maxima ≥ 5 MeV in the first 4 layers, 50 mm apart
deeper maxima: one shared region

↓

hits join their most energetic neighbour's region
photon cores separate; deep showers stay whole

↓

refit on D2 / D3, then the Λ pairing
photon 4.50·E^0.848, neutron 6.21·E^0.740

Photons shower early and the neutron late: splitting only at shallow cores frees the photons without breaking showers.

X8 — photon cores limited to 5 layers, neutron at its centroid

Change

- X7's photon clusters stay impure in 27 % of events; a core radius moved no hit: cores are narrow columns
- the contamination is neutron energy deeper in the photon's column: a core keeps only its first 5 layers
- the ZDC neutron sits at the energy centroid of its clusters, in the neutrals and in the pairing
- photon calibration refit for truncated cores: 3.95·E^0.938
- plateau over 4–6 layers in the scans; the centroid alone matched 471's prediction

Prediction (written before launch)

- 7.4 % mean, 14.0 % at 275 GeV per thrown (two-file scans, X7's calibration)

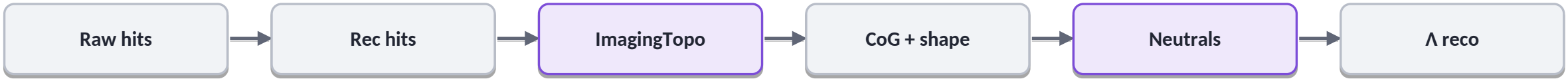
7.271 %
mean per thrown

13.970 %
at 275 GeV

current best X7_core_split: 6.853 % mean, 12.942 % at 275 GeV · ceilings 14.1 % mean, 27.8 % at 275 GeV

paired test: gained 842, lost 535, z +8.27

ACCEPTED | **fair play PASS**



purple: stages this try changes

X9 — the photons' depth as a soft term of the Λ pairing

Change

- inside the 225 mm limit, true photons still shower shallower than tagged neutron fragments
- each pair's score gains 1.0 × (depth$_1$ + depth$_2$) / 100 mm
- the pairing emulation on X8's clusters reproduces X8 within 3 events and reaches 83 % of the oracle
- weights 0.5–2 form a plateau; an energy penalty instead loses

Prediction (written before launch)

- 7.42 % mean, 14.2 % at 275 GeV per thrown (471 on X8's clusters, +2 %)

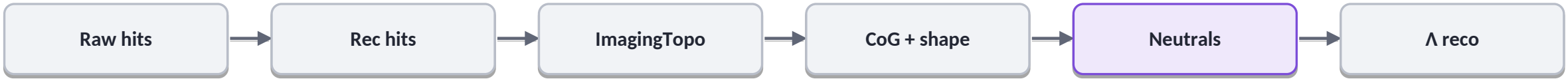
7.436 %
mean per thrown

14.101 %
at 275 GeV

current best X8_core_depth_centroid: 7.271 % mean, 13.970 % at 275 GeV · ceilings 14.1 % mean, 27.8 % at 275 GeV

paired test: gained 355, lost 160, z +8.59

ACCEPTED **fair play PASS**



purple: stages this try changes

Checkpoint 1 — held-out D4h and MSF events after X1–X5

D4h momentum	X5 matched	R1 matched	gained / lost	z	X5 % per thrown	R1 % per thrown
125 GeV	390	11	381 / 2	+19.4	0.90 %	0.03 %
175 GeV	526	41	485 / 0	+22.0	3.77 %	0.29 %
250 GeV	534	66	475 / 7	+21.3	7.76 %	0.96 %

Pooled over the three held-out momenta: gained 1341, lost 9, z +36.3. X5's per-thrown values on D4h match D4 (175 GeV 3.77 vs 3.79 %; 250 GeV 7.76 vs 7.73 %).

MSF events, 9x275 (S5, reported only)

- Λ in the acceptance found: 65 ± 10 % (R1 4 %), 23 Λ in 5000 events
- mass peak 1.121 GeV, σ 19 MeV (R1 1.149 GeV, σ 45 MeV)
- candidates 206 per 1000 events, 108 in the window: mostly not Λ

What limits next (X5, D4)

- matched 14–28 % of D4 events; wrong pair 2–4 %; mass window 2–3 %
- three clean clusters in only 28–29 % of events; ~87 % of those now convert
- photon clusters impure 32–35 %, inside the neutron's 18–21 %, merged 12 %
- next: photon-core split (X7); vertex range at the ZDC face (X6)

The gain holds on held-out events (pooled z +36); the clustering is the remaining lever.

Checkpoint 2 — held-out D4h and MSF events after X6–X9

D4h momentum	X9 matched	X5 matched	gained / lost	z	X9 % per thrown	X5 % per thrown
125 GeV	795	390	438 / 33	+18.7	1.83 %	0.90 %
175 GeV	895	526	414 / 45	+17.2	6.41 %	3.77 %
250 GeV	913	534	433 / 54	+17.2	13.26 %	7.76 %

Pooled: gained 1285, lost 132, z +30.6. X9's held-out values match D4 (250 GeV 13.26 vs 13.00 %; 175 GeV 6.42 vs 6.30 %).

MSF events, 9x275 (S5, reported only)

- Λ in the acceptance found: 96 ± 4 % (X5 65 %)
- mass peak 1.119 GeV, σ 17 MeV
- candidates 360 per 1000 events, 213 in the window: mostly not Λ

What limits next (X9, D4)

- per photon: 22 % swallowed by the neutron's cluster, 7 % by the other photon
- pairing at 83 % of the oracle; shape terms add nothing
- photon energy resolution 18–35 %: next, hit-level weights fitted on D2 / D3

X6–X9 hold on held-out events (pooled z +31 against X5); the MSF candidate rate is the physics caveat.

Predictions against results — the nine tries

try	predicted, mean / 275 GeV	measured, mean / 275 GeV	verdict	reading
X1	1.5 / 3.0 %	1.970 % / 4.309 %	ACCEPTED	beat
X2	2.84 / 5.45 %	2.913 % / 5.500 %	ACCEPTED	met
X3	3.24 / 5.98 %	3.236 % / 5.957 %	ACCEPTED	met
X4	4.31 / 7.96 %	3.278 % / 6.055 %	ACCEPTED	failed: limit set from the ZDC centre
X5	4.31 / 7.96 %	4.373 % / 8.030 %	ACCEPTED	met
X6	4.40 / 8.1 %	4.370 % / 7.997 %	REJECTED	no effect
X7	6.4 / 11.8 %	6.853 % / 12.942 %	ACCEPTED	beat
X8	7.4 / 14.0 %	7.271 % / 13.970 %	ACCEPTED	close
X9	7.42 / 14.2 %	7.436 % / 14.101 %	ACCEPTED	met

Pairing predictions from the 471 emulation held within 3 %; two-file clustering scans underestimated.

X9 losses per D4 event and the detector limits

D4 momentum	matched	clustering	wrong pair	window
100 GeV	0.37	0.52	0.08	0.04
150 GeV	0.43	0.45	0.09	0.03
200 GeV	0.46	0.41	0.10	0.03
250 GeV	0.45	0.42	0.09	0.03
275 GeV	0.43	0.43	0.10	0.04

Fractions of D4 events, first category that applies (470). Clustering: a daughter's best cluster below purity 0.8, or two daughters sharing one.

What sets the limit

- Λ mass width 14.3 MeV; 8.5 MeV with true photon energies, 14.1 MeV with the true neutron energy
- photon resolution 21–33 %, set by the 2.7 % sampling fraction
- swallowed photons sit a median 21.5 mm from the neutron, under one 24.4 mm subcell
- pairing at 83 % of its oracle
- left: crystal-section photon energy (~ +2 %), pairing retune (~ +1 %)

A larger gain needs a finer or deeper EM measurement than the Hcal sampling.

ZDC clusters after the core split — purer photons, fragmented showers

at 275 GeV (D4, 410 tables)	X5 (before)	X9 (final)
photons with best-cluster purity ≥ 0.8	0.46	0.69
photons with containment ≥ 0.5	0.49	0.27
neutron best-cluster containment, median	0.33	0.28
S0all per thrown, mean (headline.csv)	0.270 %	0.057 %

MSF 9x275 candidates per 1000 events

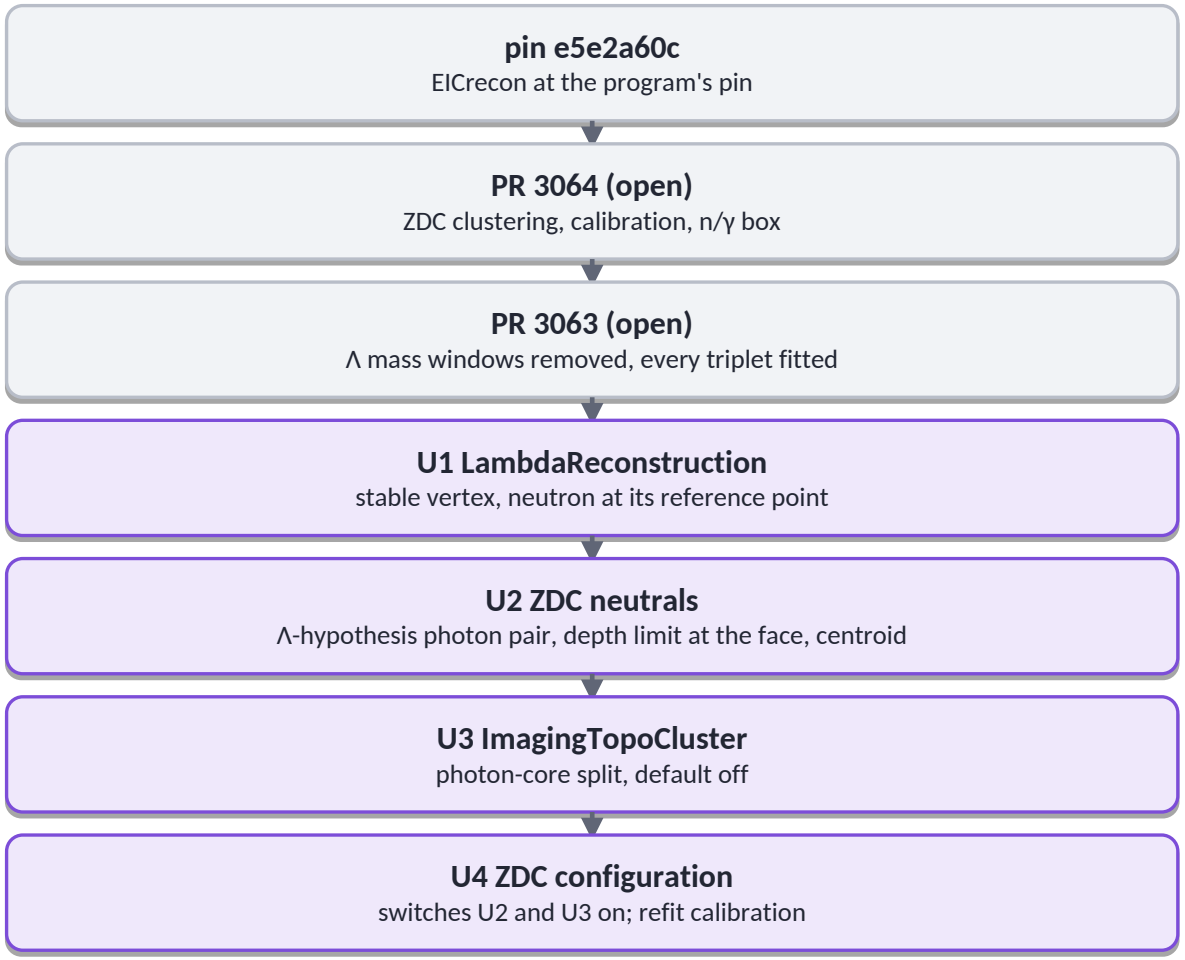
- R1 21 · X5 206 · X9 360 (213 in the window)
- X9 finds 96 ± 4 % of the 23 Λ in the acceptance
- most candidates are not Λ

What this means for other users

- a ZDC cluster is no longer one shower: S0 (purity and containment) falls while the Λ chain rises
- the neutrals factory re-sums the neutron; the photon law is refit for 5-layer cores
- the core split is off by default, so BEMC and FHCAL clustering is unchanged — argued from the code, not measured

The gun score was the only selector; cluster quality and the MSF candidate rate need their own ruler.

Upstream pieces — four patches beyond PRs 3064 and 3063



Open before any pull request

- clang-format: 40 lines over 100 columns
- one shared vertex helper; edm4hep::utils for vectors
- units on every length and energy; no magic numbers
- pairing default off in the Config struct
- Catch2 tests for the split and the pairing
- measure the cost; check the other detectors

Applying 3064, 3063, U1–U4 to the pin reproduces the branch tip a15b0db6 byte for byte. Patches: space/notes/patches/upstream/; drafts: space/reports/phase4_upstream.md. Nothing posted or pushed.

U1 fits as a review comment on PR 3063; the depth-limit convention is a one-line bug report.

Next phase — Λ purity on MSF events, then upstream

Ruler

- score Λ purity × efficiency on MSF events beside S4a_tm
- a larger MSF sample from the farm (23 Λ in 5000 events now)
- a guard on the other detectors' collections before upstream
- keep the slim template and the fair-play checker

Work

- a candidate-quality cut: fitted π^0 and Λ masses, decay depth, photon depth
- the upstream checklist on U1–U4
- small gun items: crystal-section photon energy, pairing retune
- software compensation fitted on D2 / D3

Process lessons

- JANA lengths are in cm: write them so in every config
- check where a geometry constant sits (here the ZDC centre) before using it as an offset

The gun score is near what these hits allow; the physics question now is the fake-Λ rate.